MSML640 Final Project Proposal

Weather Condition Classification from Outdoor Photographs

Image Classification with Transfer Learning

Team

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Application

We will build an image classifier that identifies the prevailing weather condition in an outdoor photograph across five classes **clear**, **cloudy**, **rainy**, **foggy**, and **snowy**. Reliable visual weather recognition is a building block for downstream systems such as dashcam-based driver assistance, autonomous-driving perception triggers, agricultural monitoring, and automatic photo organization. The task is well-suited to the four-configuration study mandated by the project: weather classes correspond directly to physical perturbations of an image (rain $\approx$ streaking and blur, fog $\approx$ low contrast and occlusion, overcast $\approx$ reduced brightness), which lets us frame a precise hypothesis rather than reporting accuracy in isolation.

Approach & Backbone

We will fine-tune an ImageNet-pretrained **EfficientNet-B0** ($\sim$5.3M parameters). EfficientNet-B0 offers a strong accuracy-per-parameter ratio for small fine-tuning datasets, trains comfortably on a single Colab T4 GPU, and is a standard transfer-learning baseline in the literature, which makes its behaviour easy to interpret. ResNet-50 will be kept as a fallback if optimization on our dataset is unstable. Following the project specification, we will train and evaluate four configurations: (1) baseline our data only; (2) our data + standard augmentation (horizontal flip, random crop, color jitter, rotation); (3) our data + synthesized images; (4) our data + synthesis + augmentation.

Dataset

Source. A combined in-the-wild dataset assembled from two streams. (a) A seed pool drawn from publicly available weather datasets (e.g., the Multi-class Weather Dataset and RFS Weather collection) to bootstrap class coverage. (b) Self-captured photographs taken across the DMV area (College Park, Lanham, Washington DC) over the next week to satisfy the in-the-wild requirement and to introduce a deliberate distribution shift between the seed pool and our test set.

Size and splits. Target $\sim$400 images total ($\sim$80 per class), with stratified splits of **70% train / 15% validation / 15% test** and no overlap between splits. All self-captured images will be reserved for validation and test where possible to stress generalization. Class imbalance, EXIF metadata, and capture-condition counts will be reported in the final writeup.

Synthesized data (Config 3 & 4). Generated with Stable Diffusion using class-specific prompts (e.g., *"rural highway in heavy fog at dusk, photographic"*, *"snowy suburban street, overcast sky"*). Synthetic images will be tagged separately so we can run clean ablations and quantify how much synthesis contributes vs. how much it distorts the data distribution.

Evaluation & Central Hypothesis

For all four configs we will report per-epoch train/val loss and accuracy curves, and a test-set confusion matrix. We will then compare the baseline directly against our best configuration with side-by-side confusion matrices and per-class error analysis (which classes fail, which images are misclassified, and why).

Robustness suite. We will evaluate the best model under four synthetic perturbations applied to the clean test set: Gaussian blur (rain proxy), brightness reduction (overcast proxy), random rectangular occlusion (fog/obstruction proxy), and additive Gaussian noise (sensor degradation). Each perturbation is implemented with documented parameters in our pipeline so results are reproducible.

Hypothesis. Augmentation will improve robustness most on perturbations that resemble the augmentation distribution itself (color jitter $\leftrightarrow$ overcast, blur in augmentation $\leftrightarrow$ rain), while synthesis will help most on minority classes where real samples are scarce. Evidence for or against this hypothesis not the absolute accuracy number is the primary deliverable of our analysis.

Proposed Timeline

Window	Milestone
Apr 24 – Apr 26	Dataset assembly: seed pool + self-captured images, cleaning, splits
Apr 27 – Apr 29	Config 1 (baseline) and Config 2 (augmentation) training + logging
Apr 30 – May 1	Synthesis pipeline; Config 3 and Config 4 training
May 2 – May 3	Robustness suite, confusion matrix analysis, slide deck, video
May 4	Final submission on ELMS + GitHub

AI Usage Disclosure

Per project policy, no AI tool was used to generate the core CV pipeline logic or our experimental design. AI assistance was used only for drafting and formatting this proposal document.